# *Web Accessibility Polices of Higher Education Institutions*

Patricia Acosta-Vargas
Facultad de Formación General
Universidad de Las Américas - UDLA
Quito, Ecuador
patricia.acosta@udla.edu.ec

Sergio Luján-Mora
Department of Software and Computing Systems
University of Alicante
Alicante, Spain
sergio.lujan@ua.es

Luis Salvador-Ullauri
Centro de Educación Continua
Escuela Politécnica Nacional - EPN
Quito, Ecuador
luis.salvador@epn.edu.ec

*Abstract* — **This article describes a study on the assessment process of web accessibility policies adopted by 51 university websites across the world, which are ranked by Webometrics. The web accessibility assessment was aimed at validating compliance with Web Content Accessibility Guidelines 2.0; these guidelines are defined and published by the World Wide Web Consortium. The primary goal of this study is to identify a group of universities with academic prestige and determine their level of commitment to providing accessible information about university facts and activities through their main websites, as well as making it available to everyone including people with disabilities.**

**Awareness about the scope of web accessibility policies is a topic of interest for organizations that are operating a website. Some policies can be applied to address internet issues such as privacy of personal data and web accessibility for people with disabilities. Policies of web accessibility are dynamic in nature due to the evolution of technology; therefore, any gaps in web accessibility should be addressed immediately and applied at different levels within an organization.**

**From a legal perspective, in some websites, policies are depicted in the legal terms posted in the website, and in the functionality and features that the website offers to its users.**

**In the evaluation phase various parameters were tested such as the implementation of web accessibility policies, version, level, availability of contact information for complaints, conformance logo, educational information, and help and support tools.**

***Keywords— accessibility, disability, policies, ranking of universities, usability, WCAG 2.0, W3C, web content accessibility, webometrics.***

## I. INTRODUCTION

Today, the World Wide Web is used in many different ways, from searches on a variety of topics to accessing online services such as those offered by higher education institutions. The number of education websites has increased greatly in recent years because authorities have recognized the many benefits that online services may provide. This success has been based on Internet availability and related technologies.

This paper presents an analysis of the Web Accessibility Polices of Higher Education Institutions following the Web Content Accessibility Guidelines 2.0 (WCAG 2.0) [1]. The analysis was applied to websites from 51 education institutions with the best ranking of webometrics. This preliminary study will alert education institutions to comply with accessibility guidelines and modify the design of their websites.

On the other hand, there are laws and regulations exist to protect the rights of vulnerable people to access many different formats of information technology; however, in practice this is not always the case due to the lack of supervision and control. In this article, the results show that high scores in web accessibility are not necessarily directly proportional to the university ranking position. By performing the evaluation of the policies for web accessibility applied in university websites, we have identified the existence of certain limitations still present for a large number of users. This article contributes significantly in the development process of more accessible websites, and awareness among web developers of the importance of websites integrating policies for web accessibility.

Web accessibility promotes inclusive education, especially the initiative proposed by UNESCO, Education for All (EFA), a global commitment to provide quality basic education to all children, youth and adults [2]. Access for all is considered to be a primary condition. Information and Communication Technologies accessibility, inclusive society, and digital gap are terms considered in the global efforts to decrease segregation of information technology accessibility among individuals with disabilities.

Accessibility enables the general public, and especially those with disabilities, to access a product or service. Web accessibility refers to the ease with which people can perceive, understand, navigate and interact with the web [1]. This article presents a deep analysis of web accessibility policies defined by the recommendation of the Web Content Accessibility Guidelines 2.0. The assessment of the web accessibility policies was completed to verify the level of

978-1-5386-3968-9/17/$31.00 ©2017 IEEE
Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

compliance of the selected universities with accessibility guidelines.

Nowadays, organizations that operate websites are putting special attention on web accessibility policies, including policies that are relevant to the area of web security issues such as privacy of personal data, and accessibility for the disabled; proper implementation of these policies should be considered which represents a practical challenge for operators of websites, therefore it is a big challenge for future research [6].

The web has become an important tool in the education field providing access information and communications globally in real-time. Some laws have been specifically created to mandate universities institutions to have websites available and accessible for students with disabilities (e.g., the Americans with Disabilities Act) [18]. Web accessibility policies establish a roadmap for universities to provide a general guide and action plan for staff members that serve as website developers within an organization [7].

There are some former accessibility policy studies carried out in South Korea including mobile accessibility and media accessibility for building an inclusive society [3].

Regarding to the article related with the accessibility of websites from Turkish universities [22], the results from assessing the level of accessibility to the examined web sites, show that a large number of websites have accessibility problems. The study suggests that it would be a good practice to adopt the principle of simple web design, using open source and management systems of free content.

The study [12] is about a longitudinal evaluation of accessibility in higher education web sites; which indicates that there are practical implications such as studies that can lead to improved guidelines, policies, and overall awareness of web accessibility for persons with disabilities.

The article "Evaluation of the web accessibility of higher-education websites" [13] describes a study to assess the accessibility of the contents concerning the websites of 20 universities from all around the world. The accessibility assessment was carried out to verify compliance with the WCAG 2.0 published by the World Wide Web Consortium (W3C). In the results, the majority of the tested websites do not achieve an acceptable level of compliance.

Regarding related works, the study [14] describes the websites of prestigious universities worldwide that accomplished few accessibility guidelines; therefore, their pages are hardly accessible.

Another research [15] confirms that there are also studies of web accessibility for higher education sites and studies of government sites in South America. In this case, the results indicate that the websites do not provide adequate levels of accessibility.

A study of quality in government websites [16] also explores the webmaster's perception and explanation of website quality. Despite the concept of advances in communication, there are surprisingly only a few studies on how webmasters perceive, experience and explain website quality or design issues.

Studies on government websites [17], mobile websites of various countries and the experience of users while browsing suggest that the websites should be designed for all types of mobile devices and operating systems. These sites must be accessible to all, regardless of their abilities, according to WCAG 2.0.

The research [20] evaluates accessibility of 25 Malaysian ministries websites using automated tools such as WAVE and Achecker. Both tools are designed to objectively evaluate web accessibility in conformance with WCAG 2.0 and United States Rehabilitation Act 1973 (Section 508). The findings reported somewhat low compliance to web accessibility standard amongst the ministries.

In universities of South America, there is a web accessibility study from Universidad de las Fuerzas Armadas-ESPE [23]. The results indicates that the websites do not provide adequate levels of accessibility.

Nowadays, continuous research about web accessibility for people with disabilities is performed globally, with a focus on laws regarding web access in accordance with the Americans with Disabilities Act, as discussed by Microsoft and the W3C [7].

The method considered in this article consisted in the selection of the top 51 universities based on the ranking of Webometrics [8].

Based on the results obtained from the web accessibility assessment, it was observed that the majority of the tested websites have not considered or implemented web accessibility policies; the universities with greater academic prestige are not aware of the importance of web accessibility policies in order to provide accessible information about the university through their websites and reached by everyone, regardless if the user has any disability.

This article is organized as follows: In section II, web accessibility is defined. Section III is devoted to discuss about the method applied in this work. In Section IV Analysis of results are presented and discussed. Finally, in section V the work is concluded with some remarks.

## II. WEB ACCESSIBILITY

WCAG 2.0 defines how to make web content more accessible to people with disabilities. Accessibility addresses a wide range of disabilities, including but not limited to visual, auditory, physical, speech, cognitive, language, learning, and neurological disabilities. Although these guidelines provide a solution path to a wide range of accessibility barriers, they are not able to address the entire needs of people with all types, degrees, and combinations of disability. Web accessibility guidelines are also aimed at making web content more friendly and usable by older individuals with changing abilities due to aging and often

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

improve usability for users in general [2].

WCAG 2.0 is divided into four main principles, and includes 12 guidelines. The principles of WCAG 2.0 are:

1. **Perceivable**: the criteria allows the product to be perceived by people, regardless of disabilities.
2. **Operable**: the user interface components and navigation must be operable.
3. **Understandable**: the information and the operation of user interface must be understandable.
4. **Robust**: the content must be robust enough that it can be interpreted reliably by a wide variety of user agents, including assistive technologies.

The aforementioned four principles include 12 guidelines which are composed of compliance criteria; each criterion considers levels of adequacy or conformity indicating the impact of accessibility.

WCAG 2.0 are divided into four general principles, organized into 12 guidelines [21]. The principles of WCAG 2.0 are the following:

- **Level A:** This is the minimum level.
- **Level AA:** It is the middle level; the website must meet all the criteria under levels A and AA.
- **Level AAA:** The highest level, at the AAA level, the website must meet all the criteria under levels A, AA and AAA.

Similarly, for each guideline and criterion of WCAG 2.0, there are three techniques to be developed to evaluate web content: sufficiency, recommended techniques and common failures.

### *A. Conformance Claims*

Conformance is defined only for web pages; however, a conformance claim may be submitted to cover one page, a series of pages, or multiple related web pages [11].

Conformance claims are not required. Authors can conform to WCAG 2.0 [19] without making a claim. In the scenario where a conformance claim is made, then the conformance claim must include the following information:

1. Date of the claim.
2. Guidelines title, version and URL "Web Content Accessibility Guidelines 2.0 at http://www.w3.org".
3. Conformance level satisfied: (Level A, AA or AAA).
4. A succinct description of the Web pages, such as a list of URL for which the claim is made, including whether subdomains are included in the claim.
5. A list of the Web content technologies relied upon [2].

The web accessibility policy may be treated independently or integrated into other policies, such as non-discrimination or equal opportunity policies. Ideally, web accessibility should also be reflected in related documents, such as brand guidelines, coding standards, and project management frameworks; through this process, accessibility can be considered as a core feature rather than an afterthought.

## III. METHOD

For this study, the first test scenario consisted on analyzing Harvard University main web site as a reference for the rest of the analysis [7]. This university is committed with the dissemination of web accessibility policies through their home page, demonstrating compliance with WCAG 2.0 guidelines. As illustrated in Fig. 1, the main page contains a link about website accessibility.

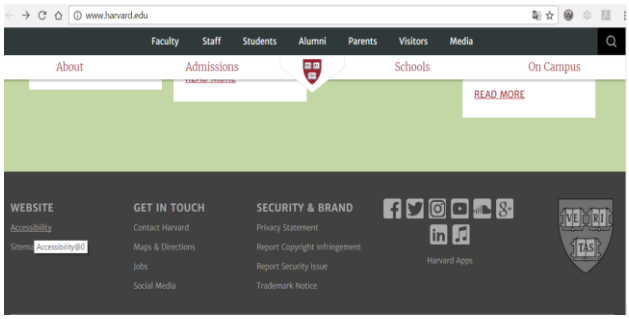



Fig. 1 Screenshot of the home page of Harvard University

Clicking on the link of web accessibility, in Fig. 2, it displays information about procedures and services for the University community and provides expertise in the development, implementation, and acquisition of standard disability-related University practices, procedures and resources.

Within the web accessibility link, as shown in Fig. 2, the user can access information about procedures and services for the university community as well as details about expertise in the development, implementation, and acquisition of standard disability-related university practices, procedures and resources at Harvard University.

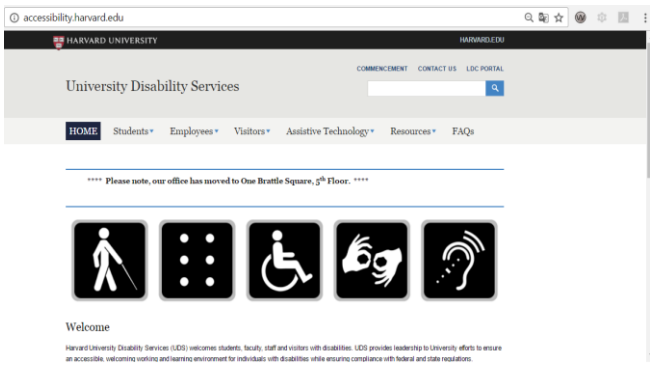



Fig. 2 Screenshot about procedures and services for the university community

For a second test scenario, as displayed in Table I, a total of 51 website samples from universities were studied based on world ranking data from Webometrics that were taken at random [8].

Table I contains the order, the name of the university, the

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

URL, and the ranking of the university.

TABLE I: UNIVERSITIES AS RANKED BY WEBOMETRICS[1], JANUARY 2017 EDITION: 2017.1.1

| No | University | URL | Ranking |
|---|---|---|---|
| 1 | Harvard University | http://www.harvard.edu/ | 1 |
| 2 | University of California Davis | http://www.ucdavis.edu/ | 31 |
| 3 | University of Melbourne | http://www.unimelb.edu.au/ | 61 |
| 4 | University of Massachusetts Amherst | http://www.umass.edu/ | 91 |
| 5 | Korea Advanced Institute of Science & Technology KAIST | http://www.kaist.ac.kr/ | 121 |
| 6 | Rheinische Friedrich Wilhelms Universität Bonn | http://www.uni-bonn.de/ | 151 |
| 7 | Universidade do Porto | http://www.up.pt/ | 181 |
| 8 | Technische Universität Wien | http://www.tuwien.ac.at/ | 211 |
| 9 | Curtin University of Technology | http://www.curtin.edu.au/ | 241 |
| 10 | San Diego State University | http://www.sdsu.edu/ | 271 |
| 11 | University of East Anglia | http://www.uea.ac.uk/ | 301 |
| 12 | University of Otago | http://www.otago.ac.nz/ | 331 |
| 13 | Yeshiva University | http://www.yu.edu/ | 361 |
| 14 | Brandeis University | http://www.brandeis.edu/ | 391 |
| 15 | University of Canterbury | http://www.canterbury.ac.nz/ | 421 |
| 16 | University of Idaho | http://www.uidaho.edu/ | 451 |
| 17 | Southern Illinois University | http://siu.edu/ | 481 |
| 18 | Indian Institute of Technology Bombay | http://www.iitb.ac.in/ | 511 |
| 19 | Università degli Studi di Trieste | http://www.units.it/ | 541 |
| 20 | Miami University of Ohio | http://miamioh.edu/ | 571 |
| 21 | Università degli Studi di Salerno | http://www.unisa.it/ | 601 |
| 22 | Brock University | http://brocku.ca/ | 631 |
| 23 | Universiti Putra Malaysia | http://www.upm.edu.my/ | 661 |
| 24 | University of Macau | http://www.umac.mo/ | 691 |
| 25 | Universidad de Alcalá | http://www.uah.es/es/ | 721 |
| 26 | Nova Southeastern University | http://www.nova.edu/ | 751 |
| 27 | Sheffield Hallam University | https://www.shu.ac.uk/ | 781 |
| 28 | Institut National des Sciences Appliquées de Lyon | https://www.insa-lyon.fr/ | 811 |
| 29 | Universidad de Costa Rica | http://www.ucr.ac.cr/ | 841 |
| 30 | Universidad de La Rioja | http://www.unirioja.es/ | 871 |
| 31 | Seconda Università degli Studi di Napoli | http://www.unina2.it/ | 901 |
| 32 | Université de Neuchâtel | http://www.unine.ch/ | 931 |
| 33 | University of Thessaly | http://www.uth.gr/ | 961 |
| 34 | Southwest Jiaotong University | http://www.swjtu.edu.cn/ | 991 |
| 35 | Universidade Federal de Santa Maria UFSM | http://site.ufsm.br/ | 1021 |
| 36 | University of Wolverhampton | http://www.wlv.ac.uk/ | 1051 |
| 37 | Wuhan University of Technology | http://www.whut.edu.cn/ | 1081 |
| 38 | Sam Houston State University | http://www.shsu.edu/ | 1111 |
| 39 | Nanjing Medical University | http://www.njmu.edu.cn/ | 1141 |
| 40 | Yamaguchi University | http://www.yamaguchi-u.ac.jp/ | 1171 |
| 41 | Yangtze University | http://www.yangtzeu.edu.cn/ | 1201 |
| 42 | Bond University | https://bond.edu.au/ | 1231 |
| 43 | Indian Institute of Technology Roorkee | http://www.iitr.ac.in/ | 1261 |
| 44 | Uludağ University | http://www.uludag.edu.tr/ | 1291 |
| 45 | College of New Jersey | http://tcnj.pages.tcnj.edu/ | 1321 |
| 46 | New York Medical College | http://www.nymc.edu/ | 1351 |
| 47 | Bard College | http://www.bard.edu/ | 1381 |
| 48 | CUNY College of Staten Island | http://www.csi.cuny.edu/ | 1411 |
| 49 | Pontificia Universidade Católica do Paraná PUCPR | http://www.pucpr.br/ | 1441 |
| 50 | Gonzaga University | http://www.gonzaga.edu/ | 1471 |
| 51 | National Chiayi University | http://www.ncyu.edu.tw/NewSite/ | 1501 |

Table II contains the information about the university ranking position, web accessibility policy standard, level of conformance, availability of the applicable policy, "Accessibility" term search within the website, Google search to validate the availability of "web accessibility site: university.edu", visual scan for accessibility option on the main website, and web accessibility support contact information; finally the total score obtained is displayed in the last column after the evaluation process.

The score was calculated as follows: for the "Version" a weight of 20 points was assigned; for "Level" 15 points; web accessibility policies implementation at the website was assigned a weight of 15 points; "Accessibility" word availability on the main page had a weight of 10 points; search for the chain "web site: university.edu" on Google if found, can have a maximum score of 15 points; if, "accessibility" is found when performing a "Visual scan" it is assigned a weight of 10 points; if the website has "Support" it is assigned 15 points. If a website complies fully with the parameters indicated by the evaluator, the site adds up to 100 points, which implies that it complies with web accessibility policies at least on its home page.

TABLE II: RANKING, ACCESSIBILITY POLICIES AND RATING

| Ranking | Version | Level | Policies | Accessibility | Google | Scan | Support | Points |
|---|---|---|---|---|---|---|---|---|
| 1 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 31 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 61 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 91 | WCAG 2.0, section 508 | A | Yes | Yes | Yes | No | Yes | 90 |
| 121 | Does not specify | No | No | Yes | Yes | No | Yes | 31 |
| 151 | WCAG 2.0, section 508 | A | Yes | Yes | Yes | Yes | Yes | 83 |
| 181 | Does not specify | No | No | Yes | Yes | No | Yes | 36 |
| 211 | Does not specify | No | No | Yes | Yes | No | Yes | 36 |
| 241 | WCAG 2.0, section 508 | AA | Yes | Yes | Yes | Yes | Yes | 88 |
| 271 | WCAG 2.0, section 508 | AA | Yes | Yes | Yes | No | Yes | 94 |
| 301 | Does not specify | No | Yes | Yes | Yes | No | Yes | 46 |
| 331 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 361 | Does not specify | No | Yes | Yes | Yes | No | Yes | 51 |
| 391 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 421 | WCAG 1.0 | No | Yes | Yes | Yes | Yes | Yes | 87 |
| 451 | WCAG, section 508 | No | Yes | Yes | Yes | Yes | Yes | 90 |
| 481 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 95 |
| 511 | Does not specify | No | Yes | Yes | Yes | No | Yes | 21 |
| 541 | Does not specify | No | Yes | Yes | Yes | Yes | Yes | 51 |
| 571 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 87 |
| 601 | WCAG 1.0, section 508 | No | Yes | Yes | Yes | Yes | Yes | 87 |
| 631 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 661 | Does not specify | No | No | Yes | Yes | No | Yes | 31 |
| 691 | Does not specify | No | No | Yes | Yes | No | Yes | 31 |
| 721 | WWC-50003 | No | Yes | Yes | Yes | Yes | Yes | 85 |
| 751 | Section 508 | No | Yes | Yes | Yes | No | Yes | 64 |
| 781 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 100 |
| 811 | Does not specify | No | No | Yes | Yes | Yes | Yes | 38 |
| 841 | Does not specify | No | No | Yes | Yes | Yes | Yes | 32 |
| 871 | Does not specify | No | No | Yes | Yes | Yes | Yes | 32 |

[1] http://www.webometrics.info/es

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 901 | Does not specify | No | No | Yes | Yes | Yes | Yes | 30 |
| 931 | Does not specify | No | No | Yes | Yes | Yes | Yes | 30 |
| 961 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 991 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1021 | Does not specify | No | No | Yes | Yes | Yes | Yes | 29 |
| 1051 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1081 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1111 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 98 |
| 1141 | Does not specify | No | No | Yes | Yes | Yes | Yes | 27 |
| 1171 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1201 | Does not specify | No | No | Yes | Yes | Yes | Yes | 27 |
| 1231 | Does not specify | No | No | Yes | Yes | Yes | Yes | 26 |
| 1261 | Does not specify | No | No | Yes | Yes | Yes | Yes | 24 |
| 1291 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1321 | WCAG 2.0 | AA | Yes | Yes | Yes | Yes | Yes | 87 |
| 1351 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1381 | WCAG, section 508 | No | Yes | Yes | Yes | Yes | Yes | 74 |
| 1411 | WCAG, section 508 | No | Yes | Yes | Yes | Yes | Yes | 73 |
| 1441 | Does not specify | No | No | Yes | Yes | Yes | Yes | 28 |
| 1471 | Section 508 | No | Yes | Yes | Yes | Yes | Yes | 74 |
| 1501 | Does not specify | No | No | Yes | Yes | Yes | Yes | 24 |

The methodology included an iterative analysis process with a subset of 51 universities as a pilot research work taking into consideration the classifications provided by Webometrics.

The iterative process applied consisted of the following steps:

**Step 1:** Select the universities, according to the global ranking of Webometrics.

**Step 2:** Visit the website of each one of the selected universities. This step is important because it allow testing of the functionality of a website.

**Step 3:** Perform a visual examination on the website of the university under evaluation. Search if there are some topics about accessibility in the page.

**Step 4:** Search on the main page if the university has a link or certificate about web accessibility.

**Step 5:** Search in each site the word "accessibility", or press the key "Ctrl + F". If this is not possible.

**Step 6:** Use the Google search engine to search for the words "web accessibility" in the entire website.

**Step 7:** Use Google to search with the pattern "web accessibility: URL of the site of the University".

**Step 8:** Then capture data the test and save.

**Step 9:** Finally, record the data in a worksheet.

## IV. Analysis of Results

In the analysis we considered 51 universities, which are presented in detail in the accompanying digital dataset[2].

Fig. 3 depicts a comparative analysis between the ranking and the score obtained in the web accessibility policies. It is observed that there is no direct relationship between ranking and web accessibility scores, meaning that not necessarily because a university is in a high ranking position it meets a high score in web accessibility.

Based on our analysis, only the following universities obtained a score of 100 points in web accessibility:

- A = Harvard University (position 1 ranked by webometrics)
- B = University of California Davis (position 31)
- C = University of Melbourne (position 61)
- D = University of Otago (position 331)
- E = Brandeis University (position 391)
- F = Brock University (position 631)
- G = Sheffield Hallam University (position 781)

The other universities considered in this study obtained a score of 24 points in web accessibility:

- I = Indian Institute of Technology Roorkee (position 1261)
- J = National Chiayi University (position 1501)

Obtained a score of 21 points in web accessibility:

- H= Indian Institute of Technology Bombay (position 511)

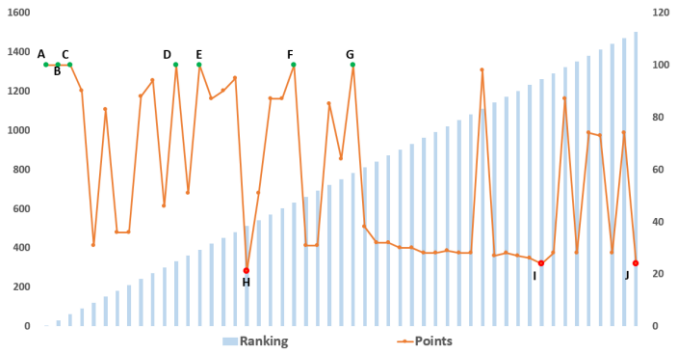



Fig. 3 Relationship between the ranking and the web accessibility policy score

Table III presents a descriptive statistic summary of the variable points in which the values obtained in the web accessibility policies were analyzed. A total of 51 samples were included in the present analysis, where the negative kurtosis indicates that there is a lower concentration of data around the mean, with a confidence level of 95%.

TABLE III: DESCRIPTIVE STATISTIC

| | | | |
|---|---|---|---|
| Average | 57.5 | Coefficient of asymmetry | 0.3 |
| Typical error | 4.3 | Rank | 79.0 |
| Median | 46.0 | Minimum | 21.0 |
| Mode | 28.0 | Maximum | 100.0 |
| Standard deviation | 30.5 | Sum | 2933.0 |
| Sample variance | 928.5 | Count | 51.0 |
| Kurtosis | -1.8 | Level of confidence (95.0%) | 8.6 |

[2] http://hdl.handle.net/10045/66227

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

In addition, the WAVE[3] tool was applied to determine the actual number of errors, alerts and number of contrast errors, which can show the behavior of university sites that are in a high ranking do not necessarily comply with adequate web accessibility policies. WAVE supports true accessibility and added as many tests for true accessibility as we can think of, including many tests for compliance issues found in the Section 508 and WCAG 2.0 guidelines. But WAVE cannot check all of the issues in these guidelines, no automated tool can do it. WAVE selects certain guidelines and standards filters in the Details panel.

In these cases, it is important the intervention of an expert in web accessibility.

Table IV presents a summary of the ranking, errors, alerts and contrast errors of WAVE tool. In which the values obtained in the web accessibility were analyzed, of which there are a total of 823 errors, 2901 alert and 1157 contrast errors.

TABLE IV: RANKING AND ERRORS OF WAVE TOOL

| Universities | Ranking | Errors | Alerts | Contrast Errors |
|---|---|---|---|---|
| Harvard University | 1 | 4 | 46 | 26 |
| University of California Davis | 31 | 20 | 88 | 12 |
| University of Melbourne | 61 | 22 | 12 | 42 |
| University of Massachusetts Amherst | 91 | 6 | 66 | 13 |
| Korea Advanced Institute of Science & Technology KAIST | 121 | 0 | 46 | 6 |
| Rheinische Friedrich Wilhelms Universität Bonn | 151 | 19 | 56 | 38 |
| Universidade do Porto | 181 | 1 | 1 | 0 |
| Technische Universität Wien | 211 | 2 | 34 | 7 |
| Curtin University of Technology | 241 | 1 | 4 | 10 |
| San Diego State University | 271 | 6 | 40 | 34 |
| University of East Anglia | 301 | 11 | 38 | 22 |
| University of Otago | 331 | 4 | 34 | 7 |
| Yeshiva University | 361 | 5 | 19 | 43 |
| Brandeis University | 391 | 1 | 21 | 12 |
| University of Canterbury | 421 | 42 | 60 | 15 |
| University of Idaho | 451 | 14 | 44 | 11 |
| Southern Illinois University | 481 | 12 | 22 | 17 |
| Indian Institute of Technology Bombay | 511 | 31 | 37 | 81 |
| Università degli Studi di Trieste | 541 | 40 | 42 | 35 |
| Miami University of Ohio | 571 | 18 | 153 | 3 |
| Università degli Studi di Salerno | 601 | 0 | 13 | 35 |
| Brock University | 631 | 7 | 59 | 17 |
| Universiti Putra Malaysia | 661 | 63 | 54 | 28 |
| University of Macau | 691 | 15 | 34 | 10 |
| Universidad de Alcalá | 721 | 9 | 61 | 14 |
| Nova Southeastern University | 751 | 0 | 23 | 0 |
| Sheffield Hallam University | 781 | 12 | 15 | 7 |
| Institut National des Sciences Appliquées de Lyon | 811 | 20 | 86 | 33 |
| Universidad de Costa Rica | 841 | 20 | 230 | 86 |
| Universidad de La Rioja | 871 | 7 | 16 | 2 |
| Seconda Università degli Studi di Napoli | 901 | 8 | 29 | 11 |
| Université de Neuchâtel | 931 | 17 | 14 | 63 |
| University of Thessaly | 961 | 6 | 15 | 9 |
| Southwest Jiaotong University | 991 | 30 | 29 | 20 |
| Universidade Federal de Santa Maria UFSM | 1021 | 13 | 36 | 105 |
| University of Wolverhampton | 1051 | 45 | 58 | 60 |
| Wuhan University of Technology | 1081 | 10 | 22 | 6 |
| Sam Houston State University | 1111 | 0 | 21 | 14 |
| Nanjing Medical University | 1141 | 23 | 102 | 2 |
| Yamaguchi University | 1171 | 19 | 14 | 0 |
| Yangtze University | 1201 | 21 | 27 | 40 |
| Bond University | 1231 | 24 | 75 | 10 |
| Indian Institute of Technology Roorkee | 1261 | 24 | 23 | 49 |
| Uludağ University | 1291 | 50 | 8 | 17 |
| College of New Jersey | 1321 | 6 | 11 | 33 |
| New York Medical College | 1351 | 32 | 18 | 7 |
| Bard College | 1381 | 18 | 58 | 13 |
| CUNY College of Staten Island | 1411 | 23 | 19 | 3 |
| Pontificia Universidade Católica do Paraná PUCPR | 1441 | 11 | 12 | 13 |
| Gonzaga University | 1471 | 10 | 50 | 4 |
| National Chiayi University | 1501 | 21 | 806 | 12 |
| Totality | | **823** | **2901** | **1157** |

Table V presents a summary of the ranking and errors, the data were grouped in ranges from 1 to 500, from 501 to 1000 and from 1001 to 1501, it was observed that in best ranking, errors, alerts and contrast errors has been reduced.

TABLE V: SUMMARY OF RANKING AND ERRORS

| Ranking | Errors | Alerts | Contrast Errors |
|---|---|---|---|
| 1 a 500 | 170 | 631 | 315 |
| 501 a 1000 | 303 | 910 | 454 |
| 1001 a 1501 | 350 | 1360 | 388 |
| **Totality** | **823** | **2901** | **1157** |

## V. CONCLUSIONS

The results obtained in this study confirm that 44 out of 51 universities evaluated did not reach the acceptable level of web accessibility policies.

It is imperative to include accessibility measures in the development of any website to correct all existing accessibility limitations on every web page, considering that implementation times are extensive. The results can act as a benchmark to compare accessibility features of academic websites and portals. The study findings and continuous feedback can provide insights into how to develop and improve websites to make them more accessible as well as establish best practices for all people and particularly people with disabilities.

[3] http://wave.webaim.org/

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.

It is a challenge to achieve the declaration of web accessibility in a simple way; most of the educational institutions do not place a web accessibility seal on their main page.

The web accessibility implementation processes adopted by different universities can be subjective. This research may serve as a baseline for future projects with a greater number of evaluators. The assessment can be improved, by assigning weights to each parameter under evaluation.

It is observed that most of the universities evaluated tend to apply the guidelines of the WCAG 2.0; however, the level of web accessibility has not been accredited by proper certification authorities.

When applying the WAVE tool, it is observed that it helps in the automatic evaluation; however, the intervention of an expert in web accessibility is required.

The results show that, although most of the universities have policies for accessibility to the web, most policies have serious shortcomings. The deficiencies identified in this work are serious enough to violate the accessibility legal rights for people with disabilities, representing a threat to this group, and which can generate a demand against the website depending on the laws and regulations executed in each country.

This study contributes with relevant data and can serve as a starting point for future research related to web accessibility policies.

## ACKNOWLEDGMENT

This research received support in the statistical analysis by the following team of subject matter experts: MSc. Alejandro Galvis, MSc. Guido Saltos, MSc. Galo Izquierdo, and MSc. Juan Carlos García.

## REFERENCES

[1] W3C. "Web Content Accessibility Guidelines (WCAG) 2.0". Available: https://www.w3.org/TR/WCAG20/. [Accessed 11 05 2017].

[2] UNESCO. "Education for All Movement". Available: http://www.unesco.org/new/en/education/themes/leading-the-international-agenda/education-for-all/ [Accessed 11 05 2017].

[3] B. Jung, C. Son, S. Park and J. Kim, "Analysis of ICT accessibility policy and implementation in South Korea," Information and Communication Technology Convergence (ICTC), Jeju, 2015, pp. 1294-1296.

[4] D. Kienle and H. Muller, "Research challenges in management and compliance of policies on the web," Website Evolution, 10th International Symposium, Beijing, 2008, pp. 83-92.

[5] D. Bradbard, C. Peters and Y. Caneva, "Web Accessibility Policies at Land-Grant Universities," Internet and Higher Education, vol. 13, no. 4, 2010, pp. 258-266.

[6] D. Beer, "Web Accessibility for People with Disabilities: An Introduction for Web Developers," Writing and Speaking in the Technology Professions: A Practical Guide. 1, Wiley-IEEE Press, 2003, pp.340.

[7] Harvard University. Available: http://www.harvard.edu/ [Accessed 11 05 2017].

[8] Webometrics. http://www.webometrics.info/ [Accessed 11 05 2017].

[9] S. Luján-Mora, "Web Accessibility among the Countries of the European Union: a Comparative Study," Actual Problems of Computer Science, vol. 1, no. 3, 2013, pp. 18-27.

[10] US Department of Justice, "ADA Standards for Accessible Design," https://www.ada.gov/regs2010/2010ADAStandards/2010ADAstandards.htm [Accessed 11 05 2017].

[11] K. Harper and J. DeWaters, "A Quest for website accessibility in higher education institutions," The Internet and Higher Education, vol. 11, Issues 3-4, 2008, pp. 160-164.

[12] S. Hackett, and B. Parmanto, "A longitudinal evaluation of accessibility: higher education web sites," Internet Research. vol. 15 Issue: 3, 2005, pp.281-294.

[13] P. Acosta-Vargas, S. Luján-Mora and L. Salvador-Ullauri, "Evaluation of the web accessibility of higher-education websites," 15th International Conference on Information Technology Based Higher Education and Training (ITHET), Istanbul, 2016, pp. 1-6.

[14] E.B. Fgee, H. A. Abakar and A. Elhounie, "Enhancement of Educational Institutions Dynamic Websites by Adding Security and Accessibility," Fourth International Conference on Next Generation Mobile Applications, Services and Technologies, IEEE Conference Publications, 2010, pp. 96-101.

[15] S. Luján-Mora, R. Navarrete and M. Peñafiel, "eGovernment and Web Accessibility in South America," First International Conference, Quito, 2014, pp. 77-82.

[16] S. Ølnes, "Small is beautiful or bigger is better? size of municipalities and quality of websites," European Conference on e-Government: 389-XV, Kidmore End: Kidmore End Academic Conferences International Limited, 2013, pp. 389-397.

[17] P. Giulianelli, C. Pons and C. Gonzalez, "Mobile government websites analysis," 7th Colombian, Computing Congress (CCC), Colombia, 2012, pp. 11-12.

[18] ADA. Americans with Disabilities Act. "Title II Public Services and Transportation," 2011.

[19] M. Vigo, J. Brown, and V. Conway, "Benchmarking web accessibility evaluation tools: measuring the harm of sole reliance on automated tests," 10th International Cross-Disciplinary Conference on Web Accessibility, ACM, 2013, pp.1.

[20]A. Ahmi, and R. Mohamad, "Current state of web accessibility of Malaysian ministries websites," AIP Conference Proceedings, vol. 1761, no. 1, AIP Publishing, 2016, pp. 020014.

[21] Universidad de Alicante, "Accesibilidad Web," Available: http://accesibilidadweb.dlsi.ua.es/?menu=niveles-2.0. [Access: 23 05 2017].

[22] K. Serhat, "The accessibility of university websites: the case of Turkish universities," Universal Access in the Information Society, vol. 10, no. 1, 2010, pp. 101-110.

[23] P. Acosta-Vargas, S. Luján-Mora and L.Salvador-Ullauri, "Evaluación de la accesibilidad de las páginas web de las universidades ecuatorianas,", XI Congreso de Ciencia y Tecnología, vol. 11, ISSN: 1390-4663, 2016, pp. 181-187.

Authorized licensed use limited to: The Open University. Downloaded on October 09,2025 at 22:33:16 UTC from IEEE Xplore. Restrictions apply.